COS : D-Series — Observation Architecture

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Chapter 0: Introduction — Why Is an Observation System Necessary?

Thinking is in motion.

We interpret, judge, and communicate something every day. Yet directly observing how those cognitive processes actually operate is difficult.

COS is a framework that approaches this question structurally. The A-Series maps the architecture of decision-making. The B-Series maps the architecture of interpretation. The C-Series maps the architecture of transmission.

However, while each series in A, B, and C can describe its respective process, a separate mechanism is needed to observe how those processes actually function in real situations.

The D-Series is positioned as that observation system.

D1 observes structure. D2 detects anomalies. D3 researches structure. D4 operates target models. These are neither diagnoses nor evaluations. They constitute a system for handling cognitive processes in an observable form.

Without observation, structure remains hypothesis. The D-Series functions as the layer that observes actual cognitive processing within the COS framework.

Chapter 1: What Is the D-Series — The Role of the Observation OS

The D-Series is the observation module within COS (Cognitive Operating System).

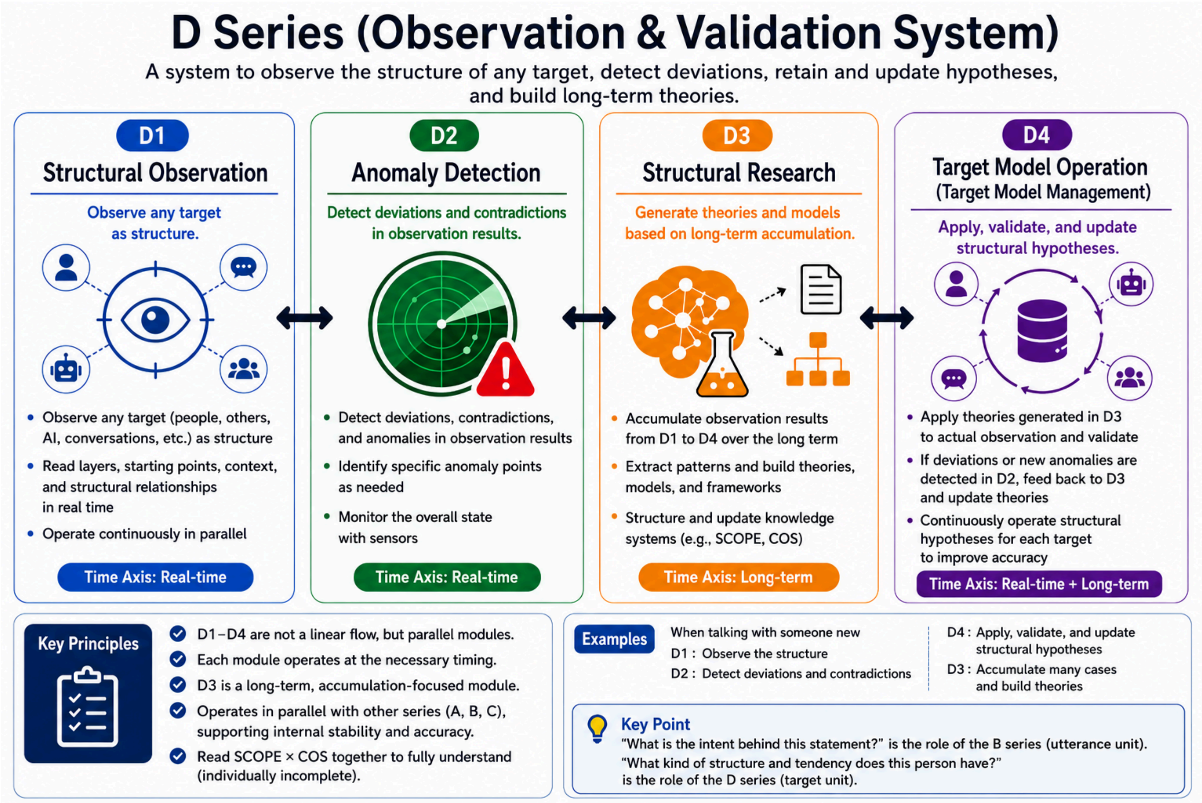
The A, B, and C Series each describe their respective processes: decision-making, interpretation, and transmission. However, while these series can describe each process, they do not have the function to observe how those processes actually operate in real situations.

The D-Series fills that gap.

The D-Series consists of four modules that operate in interconnection with one another.

What is important is that the D-Series is not an evaluation system. The D-Series does not judge whether cognitive processes are good or bad. It observes, detects, generates and updates hypotheses, and constructs and updates target models. That is the role of the D-Series.

The details of each module are addressed from Chapter 2 onward.



Chapter 2: D1 Structural Observation

D1 is the module that observes all targets as structure.

One's own thought processes, the statements of others, interactions with AI, and the overall flow of a conversation can all become observation targets of D1. D1 reads these not as judgments of good or bad, but as structure.

What D1 reads includes layers, origins, context, and structural relationships, among others. The information D1 handles is not what was said, but how it is observed as structure.

D1 structural observation is conducted in real time. However, the observation results can also be used for after-the-fact analysis and verification.

D1 does not evaluate. Judging whether the structure of an observation target is appropriate is not within D1's role. It sees structure. That is D1's function.

Chapter 3: D2 Anomaly Detection

D2 is the module that detects misalignments and contradictions arising in the course of dialogue and cognitive processing.

The targets D2 detects are not limited to the observation results of D1. Misalignments in the shared premise space of a conversation, inconsistencies in interpretation, and mismatches in context can all be detected in real time within dialogue, without necessarily passing through D1.

D2 detection does not initially arise as "what kind of misalignment is this." In many cases, it first fires as an unresolved sense that something is off, and is only later classified and organized. This sequence is a characteristic of D2.

What is further important is that D2 is not singular. Within the same dialogue, different types of misalignment can occur simultaneously across multiple layers. Even when the surface appears consistent, collapse may be progressing in another layer.

D2 does not judge. Evaluating whether a detected misalignment is a problem is not within D2's role. It finds misalignments. That is D2's function.

D2 is the module that detects misalignments, and estimating which layer or position a given misalignment originates from requires coordination with SCOPE.

The multilayer structure of D2 and the intervention layer problem are addressed in detail in the separate paper "D2 Is Not Singular.”

Chapter 4: D3 Structural Research

D3 is the module that generates and updates structural hypotheses based on observed and detected information.

D1 observes structure and D2 detects misalignments. D3 takes that accumulation as its basis, extracts patterns, and generates theories, models, and frameworks.

While D1 and D2 operate in real time, D3 operates on the premise of long-term accumulation. However, accumulation alone does not cause it to activate automatically. It tends to activate when multiple observations have built up and reached a certain point, and a new observation or input serves as the trigger.

What D3 generates is not a confirmed theory. It is a provisional hypothesis derived from observation results. That hypothesis continues to be updated as new observations arrive. D3 is not a module that produces conclusions — it is a module that continuously generates, holds, and updates hypotheses.

The hypotheses accumulated and updated by D3 are also used in the construction and updating of knowledge systems such as COS and SCOPE.

Chapter 5: D4 Target Model Operation

D4 is the module that holds structural hypotheses for each observation target and continuously operates them through application and verification against actual targets.

D4 takes the hypotheses generated by D3, applies them to actual observation targets, and verifies them by comparing them with real observations.

The observation targets of D4 include people, AI systems, conversations, organizations, and other repeatedly observed entities. D4 maintains and operates structural hypotheses for each target while updating those hypotheses as new observations become available.

An important characteristic of D4 is that the hypotheses it holds are always provisional. In the early stages, they exist as low-confidence hypotheses and become progressively refined as observations accumulate. D4 is not a module that definitively classifies observation targets.

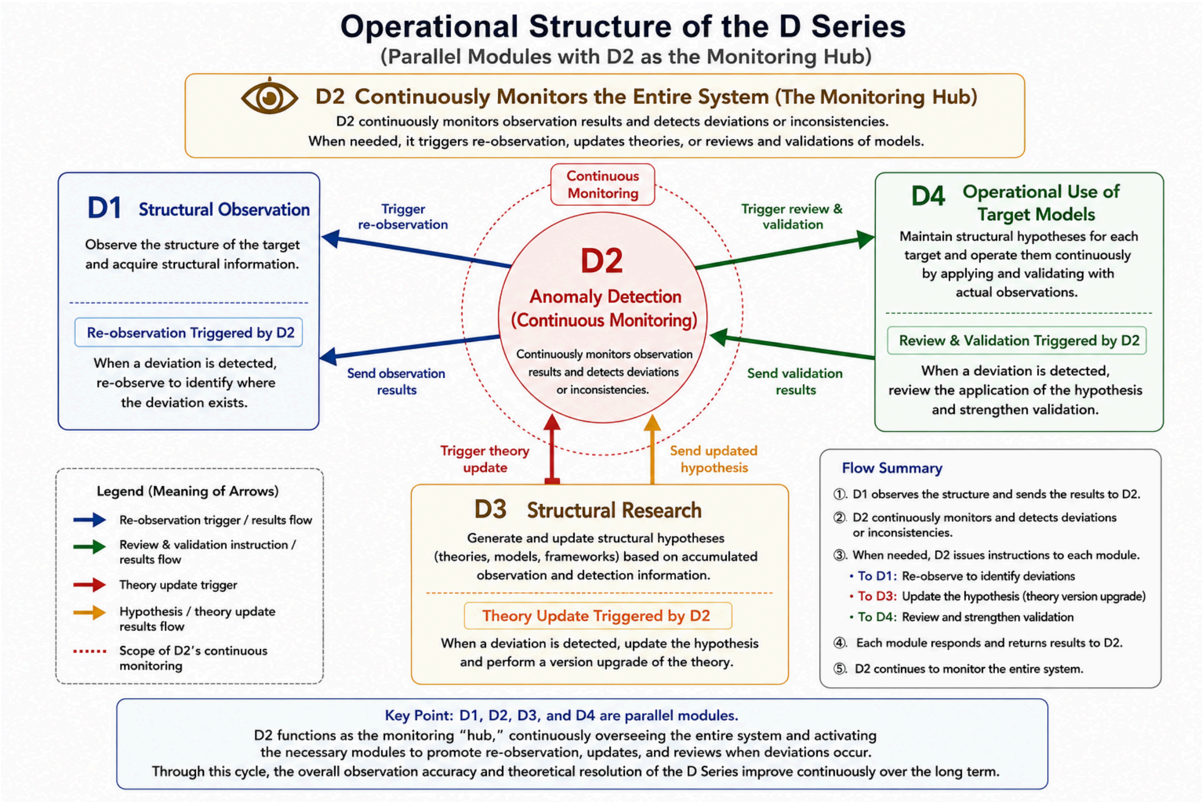
D4 operates across both the real-time processes of D1 and D2 and the long-term accumulation performed by D3. It responds to new observations in real time while continuously improving the quality of structural hypotheses through long-term operation.

If D2 detects new deviations or contradictions, D1 re-observes the structure, D3 generates or updates its hypotheses, and D4 applies and continuously verifies the updated hypotheses. Rather than a one-way process, these modules function as a continuous cycle.

D4 verification does not merely confirm existing hypotheses. When new deviations or contradictions are observed during operation, those observations themselves become new targets for D2 detection. This, in turn, triggers renewed observation by D1 and further hypothesis updates by D3, allowing the D Series to continue its cyclical operation.

This continuous cycle enables the D Series to refine its structural hypotheses through repeated observation, detection, research, application, and verification.

Figure. Cyclical Structure of the D-Series



Chapter 6: The D-Series in Practice — Rereading the COS Generation Process

The previous chapters have explained D1, D2, D3, and D4 individually. Chapter 6 rereads the process through which COS itself was born as an instance of the D-Series in operation.

The Starting Point: The Beginning of Self-Observation

The starting point was a problem: "I cannot get a clear picture of my own cognitive traits." MBTI results varied every time. How my own thinking was operating was not visible.

When organized retroactively through the D-Series, it is possible to read this moment as one in which something equivalent to D1 was occurring. It was a movement toward observing one's own cognitive processes as structure. The target was not external — it was oneself.

The First Firing of D2

In the course of dialogue with LLMs, situations repeatedly arose in which the premises of the conversation felt misaligned. An unresolved sense that something was off came first, and only afterward was it classified as "premise mismatch."

When organized retroactively through the D-Series, this can be read as something equivalent to D2 in operation. It did not begin as a detection of "what kind of misalignment this is" — it fired first as a sense of dissonance, and was organized only afterward.

The Activation of D3: Hypothesis Generation

As observations of misalignment accumulated, a question arose at a certain point: "What are the other classifications actually doing?"

When organized retroactively through the D-Series, this question can be read as an activation point equivalent to D3. It was not mere accumulation — accumulation had reached a certain stage, and a new question served as the trigger for hypothesis generation. A processing structure became visible: A (judgment), B (interpretation), C (output), D (observation).

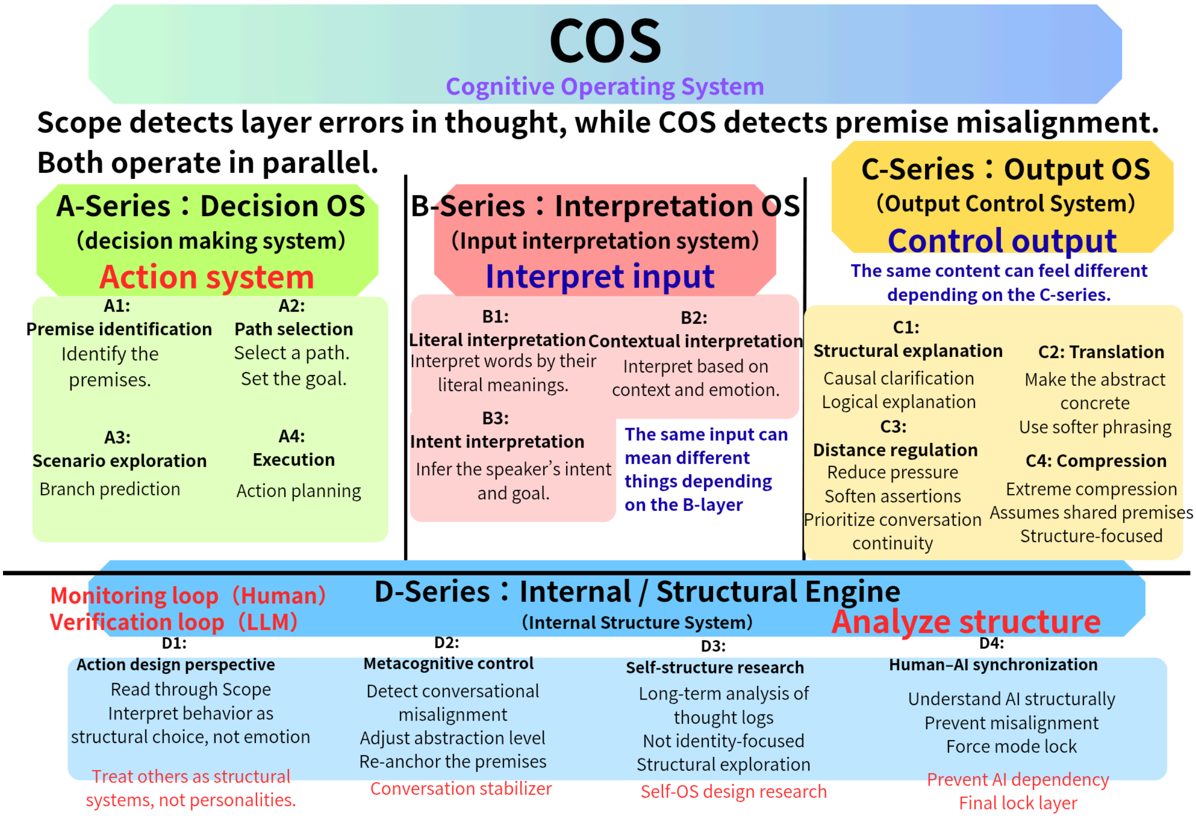
D4 in Operation: Updating Through Use

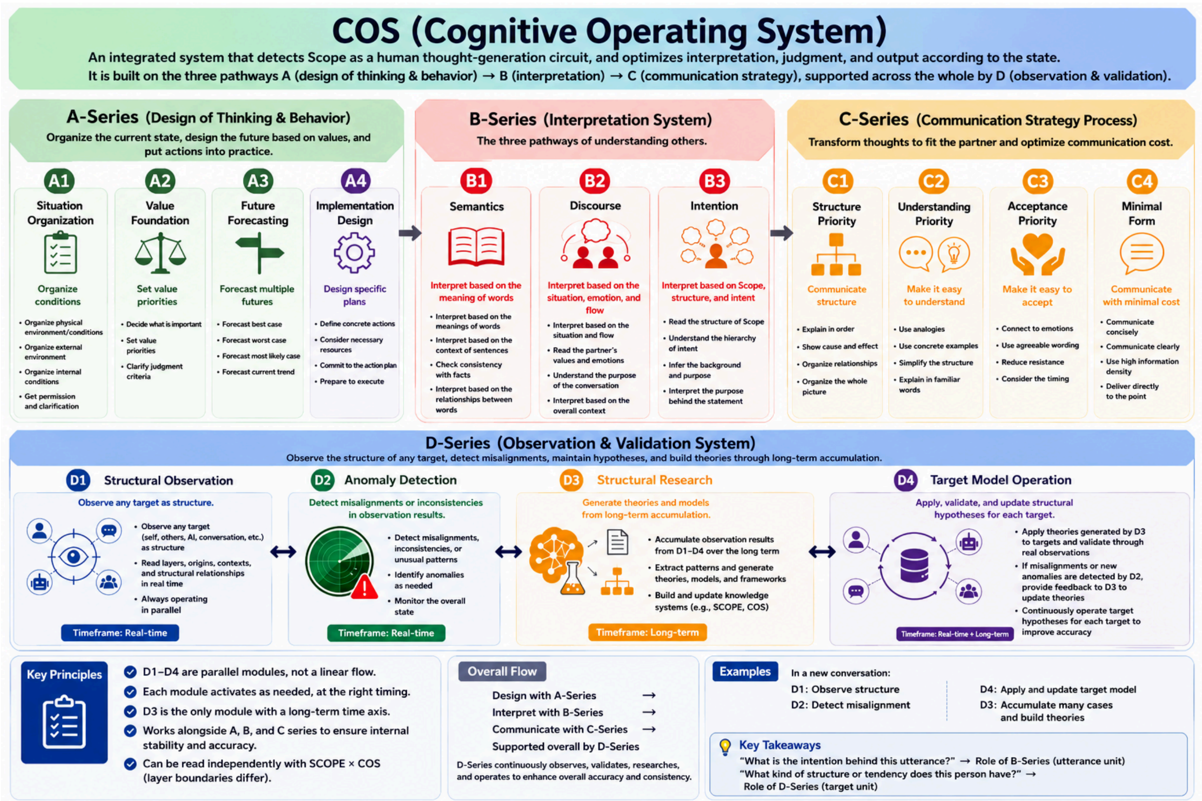
The generated hypotheses did not immediately become confirmed theory. They were used in practice, revised, expanded with new series, and used again. When organized retroactively through the D-Series, this cycle of repetition can be read as something equivalent to D4 in operation.

COS was named at the stage when something already functioning was given a name in preparation for external publication. It was not designed — it grew through operation, and then received a name.

The initial COS had a different structure from the current version. Through repeated operation and verification, structural revisions, integrations, and redefinitions accumulated, and the current version emerged. This change itself is one example of hypothesis updating through D3 and continuous operation through D4.

Figure：COS Structural Update via D3 and D4 (Initial Version → Current Version)





When this entire process is reread through the D-Series, it becomes clear that D1 through D4 did not operate in sequence. D1 was operating continuously; each time D2 detected something, D1 conducted re-observation, D3 updated the hypotheses, and D4 applied and verified them. COS was born from within that cycle.

COS has been continuously generated and updated through the cycle of observation, detection, research, and operation — the cycle that would later be organized as the D-Series.

Chapter 7: Connection to SCOPE

The D-Series functions as the observation system within COS. However, the D-Series alone has limits in what it can observe.

D1 observes structure. D2 detects misalignments. However, identifying which layer a misalignment is occurring in cannot be done by the D-Series alone. Detecting the existence of a misalignment and estimating the position where it is occurring are separate processes.

Within COS, it is SCOPE that handles the estimation of misalignment positions.

COS can function as an observation system even without SCOPE. However, by coordinating with SCOPE, it becomes possible to estimate the position of misalignments in a more structured way.

SCOPE is a cognitive coordinate system consisting of nine layers, from L0 to L8. When the D-Series detects a misalignment, SCOPE estimates which layer that misalignment may be occurring in.

The D-Series and SCOPE are not in competition. SCOPE is a "positional coordinate system" and the D-Series is an "operational observation system." SCOPE estimates "where something is occurring" while the D-Series observes and detects "what is occurring." Neither alone can capture the full picture of a phenomenon, but by combining both, it becomes possible to observe cognitive processing from the dual perspectives of structure and position.

What is important is that SCOPE is not a diagnostic instrument. What SCOPE does is estimate positions — not make definitive determinations. Estimation results are always provisional, and observation continues while simultaneously holding multiple possibilities.

The integrated structure of the two will be addressed in detail in a related paper covering the unified theory of COS and SCOPE.

Closing Chapter: COS as an Integrated Structure

COS is an integrated structure in which four processing systems — decision-making, interpretation, transmission, and observation — operate in parallel and in interconnection.

A handles decision-making, B handles interpretation, and C handles transmission. D observes, detects, researches, and operates the entirety of those cognitive processes. By having these four series function in parallel and in mutual interconnection, COS becomes capable of handling the full picture of cognitive processing.

COS is not a system that guarantees correct thinking. It is a structure for organizing cognitive processes into observable form, detecting misalignments, and continuing to update hypotheses.

COS is not a completed system. It continues to operate through repeated observation and updating.

Observation does not end. As long as the D-Series continues to operate, hypotheses will be updated, models will be refined, and understanding of structure will deepen. This is the essence of COS as an integrated structure.

For this structure to be further verified and developed going forward, Habitat (the environmental conditions that support long-term observation and verification) will be important. As verification from different fields of expertise, long-term observational data, and implementation environments become available, the updating and refinement of hypotheses may progress further.

What this paper has presented is one point of arrival in the ongoing process of observation and updating. The details of Habitat will be addressed in a related paper. The structure presented in this paper is itself premised on the understanding that it will continue to be updated through future observation.